

Preregistration Amendment v2.1 – Riemann Heat-Flow Instrument Repair

Amends: Preregistration v2.0, DOI 10.5281/zenodo.23120530

(deposited 2026-10-03, before any $t=0$ computation).

Author: Abby Davis (independent researcher, Tucson AZ). **Date:** 2026-10-03. **Status:** This amendment is timestamped on Zenodo ****before any v2.1 $t=0$ computation****. No v2.1 tracking, event detection, or Phase-1/Phase-2

computation has been run. All $t=0$ results reported herein are from v2.0

gate diagnostics (already red-flagged) or from $t=0$ instrument

characterization, which is not barred.

§1. Why v2.1 exists: the v2.0 gate outcome

On 2026-10-03 the v2.0 gate suite ran. Outcome:

Gate	Result	Measured vs threshold
V1 ($J=256 \times 512$)	FAIL	$3.79e-22$ at $z=275$ vs $< 1e-35$
V2 (first five zeros)	PASS	–
V3 (PDE residual)	FAIL	$1.25e-19$ vs $< 1e-20$
V4 (synthetic)	FAIL (marginal)	PDE sub-check $1.01e-20$ vs $< 1e-20$
V6 (null-window detector)	PASS	–
CAL-(i) ($N_{\text{off}}(0)=0$)	PASS	–
DH case ($G1/G2/G3$)	DROPPED	$G3$ spread $3.2e-07$ vs $< 1e-25$

Additionally, $t=0$ seeding found only **60 of ~104** expected zeros in

$|z| \approx 300$, and Newton polishing converged to roots displaced up to ~5

units from known references in the band $159 < |z| < 258$.

Per prereg v2.0 §7, Phase 1 and Phase 2 never ran. No event-map science data was collected. All failures were red-flagged in the lab notebook.

This amendment repairs the instrument; it does not rescue, reinterpret, or narrow any v2.0 result.

§2. Root-cause diagnosis (all verified by direct measurement)

Cause A “arithmetic precision (dps=50). At `dps=50`, the quadrature suffers ~35-digit cancellation in z-dependent high-z bands: `H_0` goes “ghost-flat” ($|H'| \sim 1e-26$), and Newton cannot polish. Raising to `dps=80` alone did **not** repair seeding (100/104 found, 22 roots displaced > 1 unit) “precision was necessary but not sufficient.

Cause B “quadrature truncation (umax=3.0). With `umax=3.0`, the fixed Gauss–Legendre nodes are too sparse to resolve `cos(zu)` at high `z`: measured $J=128 \times 256$ differences reached $2e-03$ at $z=300$. The $\hat{I}'$ mass is concentrated in $u \in [0, 1]$ ($\hat{I}'(1) \sim 1e-74$); spreading nodes over $[0, 3]$ wastes resolution where the integrand oscillates fastest. **Cause C** “the PDE check measured itself. The v2.0 `pde_residual` used a central finite difference with `dt=1e-6`. Its $O(dt^2)$ truncation is $1.25e-19$ “larger than the v2.0 V3 threshold ($1e-20$) and ~23 orders above the true PDE residual (~ $1e-42$ estimated). V3 and the V4 PDE sub-check were therefore testing finite-difference truncation, not the heat equation. The failures were test artifacts, not instrument failures.

Cause D (DH only) “separate issue. The DH G3 spread ($3.2e-07$) is a DH-quadrature matter distinct from Causes A–C. It is out of scope for this repair; the DH case remains DROPPED (§4).

§3. The v2.1 repaired instrument

Parameter	v2.0	v2.1	Why
<code>dps</code>	50	80	Cause A: rounding now at $1e-75$
<code>umax</code>	3.0	1.2	Cause B: nodes where $\hat{I}'$ has mass
<code>J</code> (production / refinement)	256 / 512	256 / 512	unchanged
<code>dH/dt</code> in PDE checks	finite difference	analytic (<code>Ht</code>)	Cause C

Measured performance of the repaired instrument (all at `t=0` unless

noted, $dps=80$, $J=256$, $umax=1.2$):

- $J=256 \hat{+} 512$ difference: **$\sim 2e-71$, uniform in z** for $|z| \hat{\approx} 300$

(was $3.79e-22$ at $z=275$).

- True PDE residual (analytic dH/dt): **$\sim 1e-85$** at all 10 test

(z, t) (was measured as $1.25e-19$ by the finite-difference artifact).

- Seeding: **104/104** expected zeros recovered in $|z| \hat{\approx} 300$

(was 60/104); worst $|H_0| \sim 1e-70$; worst displacement from known

$\hat{\pm} 2 \hat{1}^3_n < 0.01$ (was ~ 5).

Â§4. The v2.1 gate suite (revised)

All v2.0 gates are re-run from scratch under v2.1. Changes:

- **V1:** threshold **$1e-60$** (was $1e-35$; miscalibrated for the wide window). Measured $2e-71 \hat{+} 11$ orders of margin. Not permissive: tightened by 25 orders relative to the v2.0 *measured* value.

- **V2:** unchanged (passed in v2.0).

- **V2b (NEW):** seeding completeness. Requires: seeded count == reference count (104 for $|z| \hat{\approx} 300$); every seeded zero within 0.01 of a known $\hat{\pm} 2 \hat{1}^3_n$; every known $\hat{\pm} 2 \hat{1}^3_n$ within 0.01 of a seeded zero; worst $|H_0| < 1e-25$. This gate directly bars the v2.0 failure mode.

- **V3:** via `pde_residual_exact` (analytic dH/dt); threshold **$1e-60$** (was $1e-20$ via finite difference). Measured $\sim 1e-85$.

- **V4:** construction and M-refinement unchanged ($< 1e-25$); PDE sub-check via `pde_residual_exact` , threshold **$1e-60$** .

- **V5:** unchanged; runs after Phase 1 (pending at gate time).

- **V6:** unchanged (passed in v2.0).

- **CAL-(i):** unchanged (passed in v2.0).

- **DH case: DROPPED** (carried from v2.0). The comparison proceeds

as ζ vs $\text{synth-}\hat{1}$, $\text{synth-}\hat{1}01$, null-window, per v2.0 Â§4d.

Phase 1 and Phase 2 remain barred until every runnable v2.1 gate passes and is logged. The decision rule (v2.0 Â§9) is unchanged.

Â§5. What is unchanged from v2.0

The scientific question, the window ($|z| \leq 300$, not narrowed), the event taxonomy (R/C/X), the genuineness protocol G-a/b/c (G-c now uses the exact PDE residual; threshold $1e-20$ retained as a loose bound), the calibration (Â§8, including the rigorous-direction synth-1 Type-C requirement), the null-window detector check, the 48-hour study box (which starts at the first v2.1 $t=0$ evaluation), the red-flag discipline, and the scope cap (methods/phenomenology only; never a proof claim; never about RH's truth value).

Â§6. Timestamping and the $t=0$ bar

This amendment is deposited on Zenodo ****before any v2.1 $t=0$ computation****. The v2.1 gate run (which performs the first v2.1 $t=0$ evaluations, inside V6) begins only after this deposit is timestamped. The v2.0 preregistration (DOI 10.5281/zenodo.23120530) stands unaltered; this amendment is a separate dated record linked to it.

Signed: Abby Davis, 2026-10-03.

Discipline: every day we try. Preregister before computing. Red-flag every step. Nulls stay in the lab notebook; only sound results see daylight.